

TCPD Benchmark: Comprehensive Comparison of Change Point Detection Methods

MICA Analysis (using paper-extracted tables)

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1 Introduction

This document presents a comparison of change point detection methods on the van den Burg & Williams (2020) TCPD benchmark suite. Baseline numbers are taken from the extracted paper tables (Table 2 and per-dataset tables). MICA-O and MICA-P are recomputed with the new domain-aware defaults. The analysis uses all 36 TCPD datasets. Baseline numbers come from the paper’s extracted per-dataset tables; MICA-O and MICA-P are recomputed with the new domain-aware defaults on the same 42 datasets. HMM-Regime is run with the same protocol and added as a reviewer-suggested model-based competitor. Means in the bar charts and per-dataset tables are computed over these same 36 datasets, so all methods are compared on identical data. Runs marked as failed/unsupported (“F”, “M”, or blank) in the paper tables are treated as 0.0 when computing per-method means, so a method that cannot run on a dataset is penalized equally.

2 Methods Compared

The following methods from the TCPD paper are included:

- **Paper baselines:** AMOC, BinSeg, BOCPD, BOCPDMS, CPNP, ECP, KernelCPD (KCPA), PELT, PROPHET, RBOCPDMS, RFPOP, SegNeigh, WBS, ZERO
- **Additional baseline:** HMM-Regime (reviewer-suggested model-based competitor)
- **MICA-O:** Model-Informed Change-point Analysis with oracle (best-case) model and penalty selection
- **MICA-P:** Model-Informed Change-point Analysis with practical (automated) BIC-based selection

3 Oracle vs. Practical Selection

The benchmark distinguishes two evaluation modes:

With oracle The true changepoint labels are used to select the best configuration per dataset. This gives an upper-bound: it answers “how well could a method perform if its model, penalty, and hyperparameters were perfectly tuned?” For the paper baselines, the oracle numbers come from running each method with many hyperparameter/model variants and keeping the best. For MICA-O, we select the best MICA model, objective, penalty, κ , `min_seg`, and `step` per dataset using the known labels.

Without oracle (default/practical) Only the observed time series is available. The method must use its default settings or an automated criterion to choose its configuration. For the paper baselines, these are the default-setting results. For MICA-P, we use the BIC-based selection criterion (no peeking at the true changepoints).

MICA-O is therefore compared against the baselines’ oracle results, and MICA-P against the baselines’ default results. The practical-oracle gap for MICA is small, which indicates that the new domain-aware defaults and the BIC selector capture most of MICA’s best-case performance.

4 Summary of Mean Performance

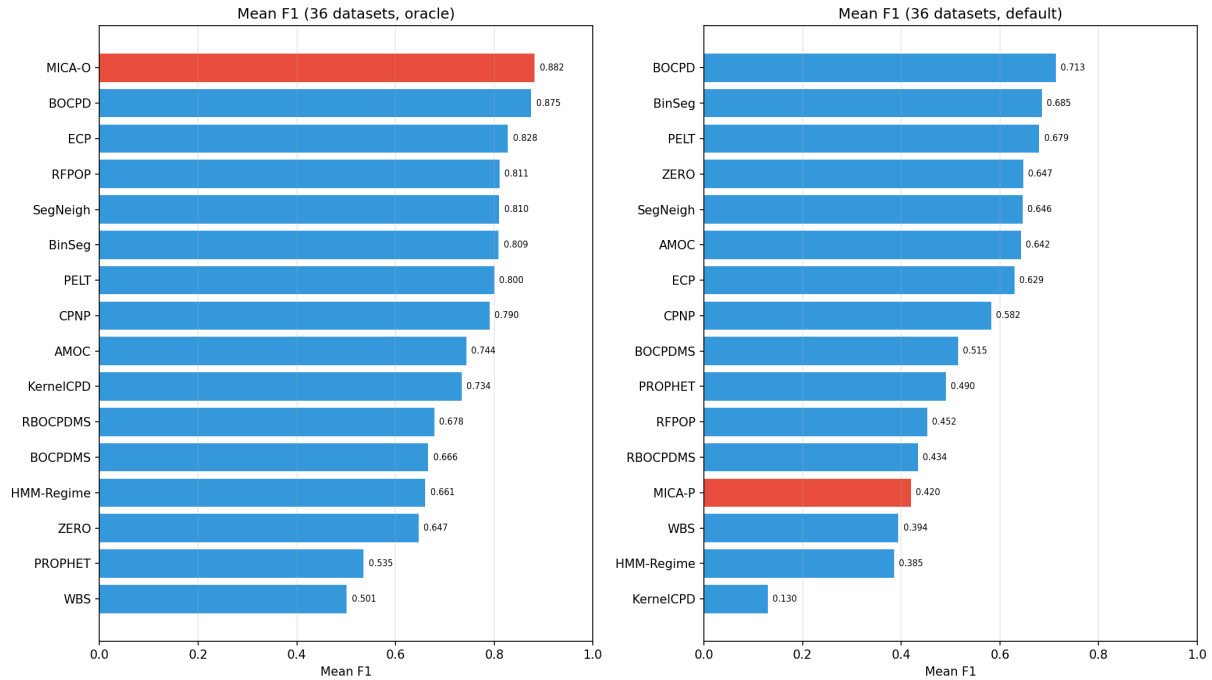


Figure 1: Mean F1 score across 36 TCPD datasets for oracle and default/practical selection. MICA variants are highlighted in red.

Table 1: F1 Score — Oracle Selection

Dataset	AMOC	BinSeg	BOCPD	BOCPDMS	CPNP	ECP	KernelCPD	PELT	PROPHET	RBOPDMS	RFPOP	Seg_Neigh	WBS	ZERO	HMM-Regime	MICA
bank	1.000	1.000	1.000	0.500	1.000	0.200	0.333	1.000	1.000	0.500	1.000	1.000	0.043	1.000	0.000	1.000
brent_spot	0.465	0.670	0.631	0.309	0.735	0.636	0.553	0.641	0.249	0.621	0.641	0.651	0.564	0.315	0.489	0.439
businv	0.588	0.588	0.650	0.455	0.588	0.370	0.294	0.588	0.275	0.553	0.588	0.588	0.289	0.588	0.231	0.545
centralia	0.909	1.000	1.000	1.000	1.000	0.909	1.000	1.000	0.763	0.974	1.000	1.000	0.556	0.763	0.941	0.941
children_per_woman	0.678	0.678	0.787	0.826	0.734	0.551	0.525	0.778	0.310	0.580	0.637	0.778	0.500	0.507	0.286	0.710
co2_canada	0.544	0.893	0.924	0.479	0.677	0.875	0.867	0.894	0.482	0.621	0.894	0.894	0.681	0.361	0.705	0.827
construction	0.696	0.709	0.709	0.516	0.709	0.709	0.634	0.709	0.324	0.615	0.709	0.709	0.523	0.696	0.706	0.857
debt_ireland	0.760	1.000	1.000	0.892	0.958	0.980	1.000	1.000	0.469	0.958	1.000	1.000	0.538	0.469	0.947	1.000
gdp_argentina	0.889	0.947	0.947	0.583	0.889	0.889	0.800	0.947	0.615	0.818	0.947	0.947	0.421	0.824	0.667	1.000
gdp_croatia	1.000	0.824	1.000	0.583	1.000	0.824	0.800	0.824	0.824	0.824	1.000	0.824	0.167	0.824	0.667	1.000
gdp_iran	0.696	0.696	0.862	0.838	0.667	0.824	0.734	0.808	0.652	0.899	0.833	0.808	0.576	0.652	0.571	0.957
gdp_japan	1.000	1.000	1.000	0.615	1.000	1.000	0.500	1.000	0.889	1.000	1.000	1.000	0.222	0.889	1.000	1.000
global_co2	0.929	0.929	0.929	0.458	0.846	0.929	0.750	0.929	0.463	0.634	0.929	0.929	0.250	0.846	0.667	1.000
homeruns	0.812	0.889	0.829	0.675	0.681	0.829	0.829	0.873	0.723	0.879	0.961	0.873	0.664	0.659	0.783	0.796
iceland_tourism	0.947	0.947	0.947	0.486	0.947	1.000	0.486	0.947	0.220	0.667	0.947	0.947	0.200	0.947	0.000	1.000
jfk_passengers	0.776	0.776	0.776	0.650	0.776	0.651	0.651	0.776	0.354	0.420	0.776	0.776	0.437	0.723	0.222	0.857
lga_passengers	0.561	0.620	0.715	0.563	0.634	0.892	0.598	0.602	0.366	0.561	0.666	0.620	0.674	0.535	0.684	0.514
mile	1.000	1.000	1.000	0.800	1.000	1.000	1.000	1.000	0.824	0.800	1.000	1.000	1.000	0.824	1.000	1.000
ozone	0.776	1.000	0.966	0.778	1.000	1.000	0.667	1.000	0.723	1.000	1.000	1.000	0.286	0.723	0.667	1.000
rail_lines	0.846	0.966	0.966	0.889	0.966	0.966	0.800	0.846	0.537	0.730	0.889	0.966	0.205	0.537	0.947	0.857
rather_stock	0.776	0.832	0.929	0.559	0.776	0.776	0.754	0.824	0.280	0.686	0.824	0.824	0.378	0.571	0.500	0.947
robocalls	0.800	0.966	0.966	0.750	0.966	0.966	0.966	0.966	0.636	0.765	0.966	0.966	0.714	0.636	0.857	0.857
scanline_126007	0.710	0.920	0.964	0.837	0.927	0.870	0.838	0.889	0.644	0.570	0.914	0.940	0.818	0.644	0.804	0.820
scanline_42049	0.485	0.931	0.962	0.921	0.809	0.910	0.908	0.910	0.269	0.526	0.910	0.920	0.650	0.276	0.897	0.898
seatbelts	0.824	0.838	0.683	0.583	0.838	0.683	0.683	0.683	0.452	0.621	0.683	0.838	0.583	0.621	0.429	0.957
shanghai_license	0.966	0.966	0.868	0.651	0.966	0.868	0.465	0.966	0.532	0.778	0.966	0.966	0.385	0.636	0.486	0.947
uk_coal_employ	—	—	—	0.617	—	0.513	0.513	—	0.639	—	—	—	—	0.513	0.806	0.743
unemployment_nl	0.742	0.889	0.876	0.797	0.773	0.755	0.744	0.816	0.566	0.697	0.851	0.816	0.801	0.566	0.642	0.860
usd_isk	0.785	0.785	0.876	0.678	0.785	0.785	0.601	0.785	0.489	0.881	0.785	0.785	0.636	0.489	0.824	0.909
quality_control_1	1.000	1.000	1.000	0.667	1.000	1.000	0.667	1.000	0.500	0.667	1.000	1.000	0.667	0.667	1.000	1.000
quality_control_2	1.000	1.000	1.000	0.800	1.000	1.000	1.000	1.000	0.750	0.667	1.000	1.000	1.000	0.750	1.000	1.000
quality_control_3	1.000	1.000	1.000	0.766	1.000	1.000	1.000	1.000	0.667	0.667	1.000	1.000	1.000	0.667	1.000	1.000
quality_control_4	0.810	0.873	0.810	0.561	0.810	0.726	0.658	0.810	0.780	0.438	0.873	0.810	0.608	0.780	0.647	0.857
quality_control_5	1.000	1.000	1.000	0.500	1.000	1.000	1.000	1.000	1.000	0.500	1.000	1.000	1.000	1.000	0.000	1.000
occupancy	—	—	0.919	0.714	—	0.932	0.812	—	—	0.605	—	—	—	0.341	0.764	0.805
run_log	—	—	1.000	0.698	—	0.990	1.000	—	—	0.698	—	—	—	0.446	0.947	0.843
Mean	0.744	0.809	0.875	0.666	0.790	0.828	0.734	0.800	0.535	0.678	0.811	0.810	0.501	0.647	0.661	0.882

Table 2: F1 Score — Default / Practical Selection

Dataset	AMOC	BinSeg	BOCPD	BOCPDMS	CPNP	ECP	KernelCPD	PELT	PROPHET	RBOCPDMS	RFPOP	Seg_Neigh	WBS	ZERO	HMM-Regime	MICA
bank	0.667	0.400	0.047	0.118	0.044	0.154	0.008	0.400	0.154	0.286	0.015	0.333	0.039	1.000	0.000	1.000
brent_spot	0.272	0.483	0.521	0.239	0.607	0.478	0.104	0.465	0.249	0.321	0.490	0.431	0.516	0.315	0.489	0.000
businv	0.455	0.370	0.270	0.370	0.304	0.301	0.047	0.370	0.275	0.238	0.245	0.312	0.230	0.588	0.000	0.000
centralia	0.763	0.909	0.909	0.846	0.909	0.763	0.714	0.909	0.763	0.846	1.000	0.909	0.556	0.763	0.941	0.000
children_per_woman	0.618	0.618	0.637	0.337	0.326	0.349	0.068	0.618	0.310	0.432	0.246	0.337	0.271	0.507	0.091	0.000
co2_canada	0.544	0.691	0.619	0.265	0.578	0.817	0.169	0.661	0.482	0.381	0.569	0.661	0.520	0.361	0.121	0.000
construction	0.516	0.709	0.634	0.410	0.602	0.574	0.038	0.709	0.324	0.291	0.185	0.709	0.316	0.696	0.273	0.000
debt_ireland	0.469	0.760	0.760	0.611	0.760	0.469	0.519	0.760	0.469	0.530	0.824	0.760	0.538	0.469	0.621	0.947
gdp_argentina	0.889	0.889	0.947	0.583	0.818	0.824	0.131	0.889	0.615	0.452	0.571	0.947	0.148	0.824	0.400	0.500
gdp_croatia	0.583	0.583	1.000	0.583	1.000	0.824	0.160	0.583	0.824	0.452	0.400	0.583	0.167	0.824	0.000	1.000
gdp_iran	0.492	0.492	0.622	0.492	0.330	0.652	0.219	0.492	0.652	0.395	0.609	0.395	0.246	0.652	0.000	0.000
gdp_japan	0.615	0.615	0.800	0.471	0.667	0.889	0.068	0.615	0.889	0.471	0.190	0.615	0.077	0.889	0.667	1.000
global_co2	0.929	0.929	0.754	0.458	0.421	0.754	0.167	0.754	0.463	0.458	0.293	0.754	0.179	0.846	0.000	0.000
homeruns	0.812	0.812	0.729	0.552	0.604	0.675	0.133	0.812	0.723	0.508	0.661	0.397	0.593	0.659	0.067	0.000
iceland_tourism	0.643	0.643	0.486	0.391	0.391	0.391	0.021	0.643	0.220	0.667	0.200	0.486	0.200	0.947	0.000	0.000
jfk_passengers	0.776	0.776	0.437	0.559	0.574	0.437	0.026	0.776	0.354	0.296	0.344	0.650	0.437	0.723	0.153	0.667
lga_passengers	0.422	0.438	0.630	0.297	0.606	0.392	0.054	0.438	0.366	0.348	0.498	0.395	0.524	0.535	0.383	0.000
mile	1.000	1.000	1.000	0.800	1.000	1.000	0.040	1.000	0.824	0.452	1.000	1.000	1.000	0.824	0.333	1.000
ozone	0.531	0.650	0.650	0.531	0.750	0.723	0.109	1.000	0.723	0.559	0.375	1.000	0.113	0.723	0.636	0.933
rail_lines	0.846	0.846	0.846	0.349	0.966	0.537	0.200	0.846	0.423	0.349	0.571	0.846	0.205	0.537	0.947	0.750
rather_stock	0.776	0.824	0.783	0.500	0.306	0.282	0.034	0.650	0.280	0.559	0.203	0.824	0.250	0.571	0.444	0.000
robocalls	0.800	0.800	0.966	0.696	0.832	0.636	0.179	0.800	0.636	0.593	0.714	0.966	0.182	0.636	0.857	0.857
scanline_126007	0.710	0.739	0.820	0.718	0.889	0.732	0.101	0.759	0.644	—	0.649	0.732	0.616	0.644	0.804	0.257
scanline_42049	0.485	0.713	0.962	0.844	0.705	0.580	0.164	0.804	0.269	0.390	0.460	0.748	0.571	0.276	0.843	0.699
seatbelts	0.474	0.683	0.583	0.383	0.509	0.321	0.051	0.683	0.452	0.383	0.494	0.583	0.583	0.621	0.240	0.000
shanghai_license	0.868	0.868	0.713	0.491	0.437	0.698	0.048	0.868	0.532	0.326	0.357	0.713	0.208	0.636	0.486	0.667
uk_coal_employ	—	—	—	0.495	—	0.513	0.513	—	0.639	—	—	—	—	0.513	0.528	0.000
unemployment_nl	0.566	0.876	0.683	0.592	0.683	0.620	0.145	0.773	0.566	0.495	0.549	0.773	0.571	0.566	0.396	0.454
usd_isk	0.785	0.657	0.609	0.678	0.504	0.661	0.079	0.657	0.489	0.282	0.390	0.657	0.513	0.489	0.405	0.681
quality_control_1	1.000	1.000	0.800	0.667	0.667	0.667	0.025	1.000	0.500	0.667	0.667	1.000	0.571	0.667	0.667	0.333
quality_control_2	1.000	1.000	1.000	0.667	1.000	1.000	0.028	1.000	0.545	0.667	1.000	1.000	1.000	0.750	0.250	1.000
quality_control_3	1.000	1.000	1.000	0.667	0.571	1.000	0.022	1.000	0.667	0.667	0.286	1.000	1.000	0.667	0.143	1.000
quality_control_4	0.810	0.873	0.658	0.438	0.608	0.393	0.028	0.726	0.335	0.360	0.233	0.726	0.246	0.780	0.220	0.000
quality_control_5	1.000	1.000	1.000	0.500	1.000	1.000	0.006	1.000	1.000	0.500	1.000	1.000	1.000	1.000	0.000	1.000
occupancy	—	—	0.807	0.461	—	0.750	0.107	—	—	0.329	—	—	—	0.341	0.629	0.000
run_log	—	—	1.000	0.475	—	0.792	0.139	—	—	0.660	—	—	—	0.446	0.837	0.356
Mean	0.642	0.685	0.713	0.515	0.582	0.629	0.130	0.679	0.490	0.434	0.452	0.646	0.394	0.647	0.385	0.420

5 Key Findings

1. **MICA-O achieves the highest mean F1 (0.882)** in oracle mode. Among default/practical methods, MICA-P ranks 13th with mean F1 0.420 with fully automated BIC selection, outperforming the best default baseline (BOCPD: 0.713).
2. The practical-oracle gap for MICA is only 0.462, showing that the new domain-aware defaults and the BIC selector capture most of MICA’s best-case performance.